

# BioSpecInfo

## Traceability Matrix — BioSpecInfo

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# Traceability Matrix — BioSpecInfo

| Field             | Value                                                                                                        |
|-------------------|--------------------------------------------------------------------------------------------------------------|
| Software          | BioSpecInfo                                                                                                  |
| Author            | Samuele Pio Provenzano                                                                                       |
| Version described | bsi-v192                                                                                                     |
| Purpose           | Connect every declared requirement to the implementation that realises it and to the bench that verifies it. |

**How to read this matrix.** Every row is a closed chain: a requirement, the point in the code that realises it, and the bench that verifies it works. A row without a bench is an **unverified** requirement and is declared as such, not hidden: the usefulness of a matrix lies precisely in making the gaps visible.

The outcomes of the benches cited here are reported in full, with their complete output, in [../evidence/RAPPORTO-VERIFICA.md](#), which is generated automatically.

## 1. Scientific requirements

| ID     | Requirement                                                                                          | Implementation                                                                                           | Verification bench                                                                                                                                                                                                                                                                                                                                                                       |
|--------|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCI-01 | A drug structure must come from an external source and match that source, not merely weigh correctly | FARM_DATA in <code>index.html</code> — <b>263 entries</b> , 36 declaring a ChEMBL identifier and formula | <code>audit_farmaci</code> — the formula is <b>rebuilt by counting the graph's atoms</b> and compared with ChEMBL's: 36 of 36. Weight alone was not enough: three pre-existing entries (lenalidomide, palbociclib, aripiprazole) were <b>isomers</b> — same weight, different molecule — and were replaced. Two ratchets: verified provenance $\geq 36$ , readable structures $\geq 240$ |
| SCI-02 | Functional groups must be recognised on the structure, not on the text of the SMILES                 | <code>bsi-spettri.js</code> §1, SMARTS patterns via RDKit                                                | <code>test_spettri</code> — 8 reference molecules, additive recognition                                                                                                                                                                                                                                                                                                                  |
| SCI-03 | A molecule with several functional groups must show them all                                         | <code>bandeDaGruppi()</code> , additive construction                                                     | <code>test_spettri</code> §1 — acetylsalicylic acid: ester <b>and</b> acid                                                                                                                                                                                                                                                                                                               |

| ID     | Requirement                                                                                                | Implementation                                                                                                        | Verification bench                                                                                                                                                                                |
|--------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCI-04 | Spectra must respect the conventions of their technique                                                    | <code>svgIR()</code> , <code>makeNMRsvg()</code> , <code>drawUVSpectrum()</code> , <code>drawMSSpectrum()</code>      | <code>test_assi</code> (SVG),<br><code>test_assi_canvas</code> (canvas)                                                                                                                           |
| SCI-05 | A computed spectrum must be reproducible                                                                   | every <code>Math.random()</code> removed from the traces                                                              | <code>test_assi</code> §3, <code>test_spettri</code> §8 — byte-for-byte comparison                                                                                                                |
| SCI-06 | The height of <sup>1</sup> H peaks must represent the integration, or not be asserted at all               | <code>makeNMRsvg()</code> ; <code>_nmrSvg()</code> draws bands of uniform height                                      | <code>test_assi</code> §4                                                                                                                                                                         |
| SCI-07 | Conjugation must act on the individual carbonyl                                                            | SMARTS <code>acidoConiug</code> , <code>estereConiug</code> , <code>chetoneConiug</code> , <code>aldeideConiug</code> | <code>test_spettri</code> §1 — aspirin 1760/1690; acetophenone 1690                                                                                                                               |
| SCI-08 | An ambiguous physical constant must not be resolved arbitrarily                                            | <code>costante_fisica</code> in <code>bsi-ai-hub.js</code> — collects every candidate                                 | <code>test_costanti</code> — 45 checks, including the refusals for ambiguity                                                                                                                      |
| SCI-09 | The limits of the predictors must be declared to the user                                                  | chart headers; <code>docs/06</code> §3.5                                                                              | <code>verifica_guida</code> — consistency between promises and code                                                                                                                               |
| SCI-10 | Molecular symmetry must determine the selection rules correctly                                            | <code>ssimm</code> section in <code>index.html</code>                                                                 | <code>test_simmetria</code> — 26 checks: normal modes, centre of inversion, mutual exclusion                                                                                                      |
| SCI-12 | A chromatogram must follow from the parameters, not be drawn by hand                                       | <code>scroma</code> section: $\sigma = t_{R} / \sqrt{N}$ , $R_s$ from distance and widths                             | <code>audit_stabilita</code> (opening and drawing), inspection of the values against the definitions                                                                                              |
| SCI-13 | A calibration line must declare the uncertainty of its coefficients and recognise the two ways it misleads | <code>staratura</code> section: ordinary least squares, F-test line $\leftrightarrow$ parabola, deleted residual      | Verified on the three example sets included: linear ( $F = 0.3$ ), non-linear ( $F = 186.6 > 7.71$ ), with an outlier (1 detected)                                                                |
| SCI-14 | A QSAR model must be evaluated on scaffolds never seen, not on analogues from the same set                 | <code>bsi-cheminfo.js</code> §9 — <code>divisionePerScaffold()</code> alongside <code>divisioneCasuale()</code>       | <code>test_cheminfo</code> — the scaffold split leaves no scaffold shared between training and test                                                                                               |
| SCI-15 | A model score must be compared against chance, not presented on its own                                    | <code>bsi-cheminfo.js</code> §9 — retraining on scrambled labels, eight repetitions                                   | <code>test_cheminfo</code> — verified in <b>both directions</b> : on a learnable signal $R^2$ 0.394 beats all eight twins (margin 0.287); on random labels $R^2$ -0.199 does <b>not</b> beat them |

| ID     | Requirement                                                                                                                                                                 | Implementation                                                                                                                                                                 | Verification bench                                                                                                                                                                                                                                                                                      |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCI-16 | Two spellings of the same molecule must not be counted twice, nor land on opposite sides of the split                                                                       | <code>bsi-cheminfo.js</code> §1 — deduplication on the canonical SMILES after the largest fragment                                                                             | <code>test_cheminfo</code> — <code>OC(=O)C</code> and <code>CC(=O)O</code> collapse into one entry; salts lose the counterion                                                                                                                                                                           |
| SCI-17 | Similarity between molecules must be computed, not estimated                                                                                                                | <code>bsi-cheminfo.js</code> §3 — Tanimoto and Dice over Morgan, RDKit, pattern and MACCS fingerprints                                                                         | <code>test_cheminfo</code> — compared against hand-computable values, including the $0/0 = 1$ convention                                                                                                                                                                                                |
| SCI-18 | A validation on a single split, over a small set, is a noisy number: it must come with its dispersion                                                                       | <code>bsi-cheminfo.js</code> §9-bis — <code>validazioneIncrociata()</code> over <b>scaffold-grouped</b> folds                                                                  | <code>test_cheminfo</code> — the folds cover every molecule once, <b>no scaffold sits in two folds</b> , and the same seed gives the same folds                                                                                                                                                         |
| SCI-19 | A result that changes on every run is verifiable by nobody                                                                                                                  | seeded mulberry32 generator for splits, scrambles and folds; the seed is written into the report                                                                               | <code>test_cheminfo</code> — two calls with the same seed give identical folds; the method report declares the seed                                                                                                                                                                                     |
| SCI-20 | Results must be able to leave the tool, and the method must leave with them                                                                                                 | <code>bsi-cheminfo.js</code> §9-ter — <code>esportaCsv()</code> and <code>rapportoMetodo()</code> , both formed in memory in the browser                                       | <code>test_cheminfo</code> — columns consistent with the header, a name containing a comma quoted, residual computed, empty cell for missing activity; the report carries seed, fingerprint, verdict, limits                                                                                            |
| SCI-21 | A number presented as structural similarity must come from a fingerprint, and one implementation must compute it                                                            | <code>rdkit_lab.html</code> loads <code>bsi-cheminfo.js</code> and uses its <code>tanimoto()</code> and <code>fingerprint()</code> ; the eight-bit similarity has been removed | <code>test_cheminfo</code> — page and engine give the same Tanimoto over all <b>28 pairs</b> , deviation <b>exactly 0</b> ; the fake-similarity functions no longer exist; every reference structure is valid and carries the <b>monoisotopic mass</b> of the drug it names                             |
| SCI-22 | Conversions between molecular languages must be checked against an external source or against hand calculation, and coordinates written to a file must be real and declared | <code>bsi-molingue.js</code> — <b>seventeen outputs</b> , from SMILES to CML                                                                                                   | <code>test_lingue</code> — 7 <b>literature InChI keys</b> , full SMILES → molfile → SMILES round-trip, 26 hand-written names, 7 <b>required refusals</b> ; mass composition against hand calculation (alanine C 40.44%, aspirin C 60.00%); XYZ and PDB with non-null coordinates <b>and</b> declared 2D |

| ID     | Requirement                                                                                                                                  | Implementation                                                                                                                                                                    | Verification bench                                                                                                                                                                                                                                                                                                        |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCI-23 | Angles, lengths and dihedrals are measured on the coordinates, and on flat coordinates they are not shown                                    | dot product for angles, cross product for dihedrals                                                                                                                               | <code>test_mol3d</code> — tetrahedron <b>109.4712°</b> , water 104.47°, CO <sub>2</sub> 180°, BF <sub>3</sub> 120°, ammonia 106.13°; dihedrals 0°, 60°, 90°, 180°; ethanol C-C 1.509 Å; and the refusal tested in both directions                                                                                         |
| SCI-24 | A spectrum read from a file must be decoded according to the format, and the peaks found must be the ones that are there — no more, no fewer | <code>bsi-spettrolettore.js</code> — JCAMP-DX with full ASDf (PMAI, DIF, DUP) and the integrity check the format prescribes; peaks by <b>prominence</b> , noise from MAD × 1.4826 | <code>test_spettrolettore</code> — <b>39 checks</b> . On three Gaussians built at 1715, 2950, 3400 cm <sup>-1</sup> under noise it finds three: 1716, 2952, 3402. <b>Both ways</b> : on pure noise it finds <b>zero</b> . ASDf verified symbol by symbol ( <code>abcdefghi</code> = -1...-9; <code>n</code> = -5, not -4) |
| SCI-25 | A band must be declared <i>compatible with</i> , never assigned to a single possibility                                                      | <code>BANDE_IR</code> — 24 ranges; <code>assegnaIR()</code> returns <b>all</b> compatible assignments                                                                             | <code>test_spettrolettore</code> — at 1715 cm <sup>-1</sup> both a ketone C=O and a carboxylic acid C=O appear; the section states that it does not deduce the structure                                                                                                                                                  |
| SCI-26 | A published model must be checked against <b>its own</b> declared error, and a learned model against chance                                  | <code>bsi-cheminfo.js</code> §10 — ESOL (Delaney 2004) with its uncertainty, four drug-likeness filters reported separately, random forest with out-of-bag error                  | <code>test_spettrolettore</code> §4-5 — ESOL against four experimental values, mean deviation <b>0.68</b> , inside the declared ~1 log unit; forest out-of-bag R <sup>2</sup> <b>0.90</b> on a non-linear relationship and <b>-0.16</b> on pure noise; deterministic for a given seed and different for another           |
| SCI-27 | A structure without a source must not get in, not even one that is known                                                                     | collected from ChEMBL, <code>tools/dati/farmac_i_v187.json</code> carries each entry's identifier                                                                                 | <b>Two refusals</b> : ivermectin ( <code>structure_type</code> NONE , a mixture of homologues) and semaglutide (a protein, <code>SEQ</code> ). And two entries kept under the <b>record's</b> name rather than the query's: «morphine» → apomorphine, «levothyroxine» → liothyronine                                      |

| ID     | Requirement                                                                                                                                       | Implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Verification bench                                                                                                                                                                                                                                                                                                                                                                                 |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCI-28 | A predicted spectrum must say WHICH atoms produce each signal, and the deviation must be measured on molecules that did not choose the parameters | <code>bsi-prettsch.js</code> — the estimation tables transcribed in full from the source (Prettsch 4th ed.: 91 $^{13}\text{C}$ and 66 $^1\text{H}$ rows for benzenes, 42 for ethylenes, 31 for alkanes, 24 for $^{13}\text{C}$ aliphatics, 4×4 steric corrections); <code>bsi-nmr.js</code> — the reasoning that applies them: ring position obtained by walking the cycle, additive scheme with a cyclic reference compound, chemical equivalence by shell environment code | <code>test_nmr</code> — <b>56 checks</b> . Two separate sets: tuning <b>0.71 ppm</b> over 9 molecules, <b>validation 0.92 ppm</b> over 22 never used for tuning, worst case <b>4.9 ppm</b> , $^1\text{H}$ <b>0.06 ppm</b> (0.03 on the aromatics alone). The bench requires validation to stay <b>worse</b> than tuning: were they equal, someone would have moved a molecule between the two sets |
| SCI-29 | Chemically equivalent atoms must give ONE signal, not one per atom                                                                                | environment code built as concentric shells by distance (the HOSE-code principle), with aromatic bonds written as such and not in Kekulé form                                                                                                                                                                                                                                                                                                                                | <code>test_nmr</code> — eight molecules whose signal count is known: benzene <b>1</b> (not six), toluene 5, p-xylene 3, naphthalene 3, biphenyl 4, aspirin 9. All correct                                                                                                                                                                                                                          |
| SCI-30 | From an image one can recover the SHAPE of a spectrum, not its calibration                                                                        | <code>bsi-spettrolettore.js</code> §6-bis — trace extraction from the pixels column by column, background estimated from the median, empty columns interpolated                                                                                                                                                                                                                                                                                                              | <code>test_spettrolettore</code> — on a <b>constructed</b> figure with Gaussians at 1715/2950/3400 $\text{cm}^{-1}$ it reads <b>1713/2947/3403</b> from the pixels alone; <b>in both directions</b> , on a blank sheet it refuses instead of inventing a trace; and it declares that the axis scale is not in the pixels                                                                           |
| SCI-31 | A two-dimensional NMR map must say WHICH atoms produce each correlation, and must not connect what the topology does not connect                  | <code>bsi-nmr2d.js</code> — one-bond HSQC with inverted sign on $\text{CH}_2$ , three-bond COSY between non-equivalent protons, two- and three-bond HMBC; exchangeables excluded from off-diagonal spots                                                                                                                                                                                                                                                                     | <code>test_nmr2d</code> — <b>40 checks</b> , in both directions: hexafluorobenzene gives no HSQC spots, benzene <b>no</b> off-diagonal COSY spots (its six protons are equivalent), methane <b>no</b> HMBC spots despite having four protons                                                                                                                                                       |
| SCI-32 | A program-built 3D geometry must carry THE SAME indices as the predictor, otherwise the peak↔atom link lights up the wrong atom                   | <code>bsi-geom3d.js</code> — distance geometry over the <code>bsi-nmr.js</code> graph, hydrogens appended at the end, generator seeded from the SMILES                                                                                                                                                                                                                                                                                                                       | <code>test_geom3d</code> — <b>29 checks</b> : every heavy atom has the same index and element across five molecules; bonds and angles within tolerance of literature values; <b>in both directions</b> benzene comes out planar and cyclohexane does not                                                                                                                                           |

| ID     | Requirement                                                                                                                                                              | Implementation                                                                                                                                                                                     | Verification bench                                                                                                                                                                                                                                                                                                                                                                         |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCI-33 | A document must be opened in full, and whatever could NOT be read must be declared — a reader that returns little without saying so makes the document look nearly empty | <code>bsi-documento.js</code> — PDF page by page with its text layer (PDF.js), images, text, Office archives opened with <code>DecompressionStream</code> and no libraries                         | <code>test_documento</code> — <b>47 checks</b> : PDF, Word and text genuinely opened and their content verified; <b>in both directions</b> an image declares there is no optical character recognition and produces no working, an unknown format is refused rather than guessed, a binary file named <code>.txt</code> is recognised as binary                                            |
| SCI-34 | Spectroscopic data recognised in a text must carry their PROVENANCE, and the working must declare what it is missing                                                     | <code>bsi-quesito.js</code> — recognition by section (IR, MS, $^1\text{H}$ , $^{13}\text{C}$ ) with the boundaries between sections, then <code>bsi-elucida.js</code> for the step-by-step working | <code>test_documento</code> — formula, bands, masses with intensities, $^1\text{H}$ signals with integral and multiplicity, $^{13}\text{C}$ signals; <b>in both directions</b> : $\text{CDCl}_3$ does not become the compound, a band outside 400-4000 is not a band, without an MS label there are no masses, a new paragraph closes the section, a text with no spectra produces no data |
| ING-01 | One function must not have two implementations: the React/FastAPI stack must run THE SAME files as the application                                                       | <code>stack/api/worker/motore.mjs</code> — a Node process loads the repository modules with RDKit WebAssembly; no Python engine                                                                    | <code>test_stack</code> — <b>15 checks</b> : compares the SHA-256 fingerprints declared by <code>GET /salute</code> with those of the files on disk, verifies the numbers match the browser's, and that no <code>.ts / .tsx</code> file contains a shift table                                                                                                                             |

## 2. AI agent requirements

| ID    | Requirement                                                                 | Implementation                                                                   | Verification bench                                      |
|-------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------|---------------------------------------------------------|
| AI-01 | A model retired by the provider must not block the assistant                | <code>modelloSuggestitoDaErrore()</code> , <code>bsi_modelli_ko</code> blacklist | <code>test_404</code> — 36 checks                       |
| AI-02 | A provider unreachable from the browser must be remembered and demoted      | <code>bsi_prov_ko</code> , 24 h TTL, <code>providerUtilizzabili()</code>         | <code>test_ko</code> (35), <code>browser_ko</code> (37) |
| AI-03 | A temporary overload must produce a retry, not an error                     | <code>erroreTemporaneo()</code> , <code>TENTATIVI_TEMPORANEO</code>              | <code>test_503</code> — 40 checks                       |
| AI-04 | An imposed wait longer than 10 s must switch provider if a ready one exists | <code>ATTESA_TROPPO_LUNGA_MS</code>                                              | <code>test_attesalunga</code> — 14 checks               |

| ID    | Requirement                                                                 | Implementation                                            | Verification bench                                                                                          |
|-------|-----------------------------------------------------------------------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| AI-05 | The provider's token ceiling must be learned from the error, not hard-coded | <code>tettoDaErrore()</code> , <code>MARGINE_TETTO</code> | <code>test_tetto</code> — 32 checks                                                                         |
| AI-06 | Opaque reasoning signatures must be returned to the provider                | capture and replay in <code>appendAgentTurn()</code>      | <code>test_firma</code> — 31 checks                                                                         |
| AI-07 | Two overlapping submissions must not corrupt the history                    | <code>_turnoInCorso</code> in <code>send()</code>         | <code>test_doppioinvio</code> — 7 checks                                                                    |
| AI-08 | The user must be able to measure which providers they can actually reach    | <code>provaTuttiIFornitori()</code>                       | <code>browser_prova</code> — 29 checks                                                                      |
| AI-09 | Paid providers must not be offered as free ones                             | separation by <code>PROVIDERS[id].free</code>             | <code>browser_prova</code> §D2                                                                              |
| AI-10 | A documented route must exist to remove CORS and rate limits                | <code>proxy/spectra-proxy.js</code> + in-app guide        | <code>test_guidaproxy</code> (12),<br><code>browser_proxy</code> (15),<br><code>browser_proxyui</code> (17) |

### 3. Stability requirements

| ID     | Requirement                                                         | Implementation                                                | Verification bench                                                        |
|--------|---------------------------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------------------------------|
| STA-01 | A long session must accumulate neither DOM nodes nor timers         | section lifecycle management                                  | <code>audit_stabilita</code> §1 — 92 sections × 5 rounds                  |
| STA-02 | Exhausting <code>localStorage</code> must not make the app unusable | guarded writes; persistent warning in the File Manager        | <code>audit_quota</code> (10),<br><code>test_filemanager</code> (15)      |
| STA-03 | No rejected promise may remain unhandled                            | systematic <code>.catch()</code>                              | <code>audit_promesse</code> — 22 checks, healthy network and dead network |
| STA-04 | Corrupted saved data must not prevent startup                       | <code>loadJSON()</code> with fallback                         | <code>audit_stabilita</code> §4                                           |
| STA-05 | A slow network must not be treated as an absent one                 | <code>ATTESA_RETE_MS = 3500</code> in <code>sw.js</code>      | <code>test_sw</code> §3 and §3-bis — measured 3,507 ms                    |
| STA-06 | An update must not destroy work in progress                         | <code>BSI_SW_UPDATED</code> message, decision taken page-side | <code>test_sw</code> §4                                                   |
| STA-07 | Browsing molecules must not exhaust the WebGL contexts              | reuse of the 3D viewer at all 5 creation points               | <code>test_visore3d</code> — from 7 constructions down to 1               |
| STA-08 | An unavailable IndexedDB store must be declared, not fatal          | <code>dbStore()</code> over a promise; persistent warning     | <code>test_filemanager</code> §2                                          |

## 4. Interface and accessibility requirements

| ID    | Requirement                                                                                                                       | Implementation                                                                                                                           | Verification bench                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UI-01 | Every page must open without JavaScript errors                                                                                    | —                                                                                                                                        | <code>audit_stabilita</code> §5 — 14 pages                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| UI-02 | Charts must be sharp on high-density screens                                                                                      | <code>bsiNitido()</code> on the canvas contexts                                                                                          | <code>audit_grafici</code> — 40 canvases                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| UI-03 | The app must work at 390 px width                                                                                                 | <code>auto-fit</code> grids, no fixed column                                                                                             | <code>audit_mobile</code> — the document's <code>scrollWidth</code> across all 92 sections at 390 px; <code>audit_stabilita</code> for errors in a phone viewport                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| UI-04 | The user must be able to know which version they are running and force the update                                                 | "Updates" entry in the ✨ panel                                                                                                           | <code>test_aggiorna</code> — 9 checks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| UI-05 | Data deletion must be selective and reversible in its choices                                                                     | <code>bsiCancellaDati()</code> by group                                                                                                  | <code>browser_reset</code> — 24 checks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| UI-08 | Surfaces that appear only after an action must also meet WCAG AA contrast                                                         | the six panels of the Cheminformatics section                                                                                            | <code>test_cheminfo</code> — the analysis is run, the panels opened in turn, <b>460 text elements measured, 0 defects</b> , with the WCAG formulas rewritten inside the bench                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| UI-09 | Opening a section must not block the page                                                                                         | 3Dmol's first <code>render()</code> moved out of the click; the 296 synthesis figures <b>and the 263 drug cards</b> drawn in 8 ms slices | <code>test_fluidita</code> — <b>92 sections traversed</b> , median switch 16 ms, worst 87 ms, <b>0 sections above 100 ms</b> (there were 3 with the new drugs, at 159 ms); plus the opposite proofs: the WebGL canvas appears anyway, all 296 figures exist shortly after, and printing does not come out mute. For the viewer the claim is <b>structural, not temporal</b> : immediately after the click handler returns, the WebGL context must not exist; shortly after it must — because a millisecond threshold was measuring <code>SwiftShader</code> rather than the code (14...192 ms on identical code) |
| UI-10 | The interface must switch among many languages, declaring how much each is translated, and carrying the writing direction with it | <code>bsi-lingue.js</code> with <b>fourteen languages</b> , a searchable selector, <code>dir="rtl"</code> for Arabic                     | <code>test_lingue</code> — <b>170 skeleton elements out of 170 in each of the thirteen languages</b> , coverage computed by applying the dictionary and counting; writing direction tested in both directions; navigation categories and the search placeholder included                                                                                                                                                                                                                                                                                                                                         |
| UI-11 | The molecule must show how it holds together, and the viewer must choose the angle                                                | <code>bsi-mol3d.js</code> : formation from dust with staggered arrival, and atom selection by click                                      | <code>test_mol3d</code> — the canvas <b>pixels</b> at 130 ms and 2.3 s, the arrival order (skeleton before hydrogens) measured on each atom's progress, and the selection of 1, 2, 3 and 4 atoms                                                                                                                                                                                                                                                                                                                                                                                                                 |

## 5. Security and privacy requirements

| ID     | Requirement                                                                          | Implementation                                                                                    | Verification bench                                                                                                                                                                                                                                                                            |
|--------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SEC-01 | No API key may be present in the repository                                          | keys only in <code>localStorage</code> or in the Worker's secrets                                 | <code>tools/verifica-sicurezza.js</code> — 362 tracked files, 8 credential shapes                                                                                                                                                                                                             |
| SEC-02 | No personal data may leave the device without an explicit action                     | local-first architecture, telemetry disabled                                                      | <code>audit_rete</code> — <b>direct verification</b> : a canary value seeded into 71 stores of the user's data, the application used across 92 sections on 6 pages, and the URL, headers and body of every request inspected. Plus <code>verifica-sicurezza</code> on the two exit mechanisms |
| SEC-03 | Passwords must not appear in clear text in the source                                | SHA-256 in <code>file_manager.html</code>                                                         | <code>tools/verifica-sicurezza.js</code>                                                                                                                                                                                                                                                      |
| SEC-07 | No credential may EVER have entered the repository, not even in a commit later fixed | no key has ever been committed; keys live in <code>localStorage</code> or in the Worker's secrets | <code>audit_storia</code> — <b>1,551 distinct versions of text files across 390 commits</b> , 9 patterns tested in both directions; fails on a shallow clone, because it would measure less surface                                                                                           |
| SEC-04 | No conflict marker may reach publication                                             | —                                                                                                 | <code>verifica_guida</code> §12 — 58 text files                                                                                                                                                                                                                                               |

**SEC-03 — declared limitation.** GitHub Pages serves static files: any page-side access control can be bypassed by whoever reads the source. The File Manager's protection is a deterrent, **not** a security control, and the documentation says so. Furthermore the password remained in clear text in the git history until its removal, and taking a secret out of the files does not take it out of the history: `git log -p` hands it to anyone. The only effective remedy was to change it, and that **has been done** at version `bsi-v188`. The old one remains in the history and no longer opens anything.

## 6. Requirements not yet verified automatically

Declared for completeness. They are covered by documentary inspection or by manual testing, and their automation is planned.

| ID             | Requirement                                                      | Current coverage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>UI-06</b>   | Full WCAG 2.1 AA conformance                                     | the mechanical part is automated across <b>13 pages and 92 sections</b> ( <code>tools/verifica-accessibilita.js</code> , 38,407 text elements: it was 19,751 before text over gradients entered the measurement, and 37,409 before the «Spectrum reader» section). <b>Of the 1,069 contrast defects that emerged, none remain: 0 measured</b> over those same 92 sections, and the value is recorded as a baseline the bench defends. Unlabelled fields are <b>zero</b> : see the box in <code>docs/09</code> §4. What remains outside is text inside SVGs and text over a real background image, counted on every run (D-09) |
| <b>PERF-01</b> | First-paint time on a low-end device                             | manual cross-device testing ( <code>docs/02</code> §4)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>SCI-11</b>  | Structures of 2 entries that are not single molecules (it was 6) | <b>not representable</b> : Ivermectin is a mixture of homologues, Coartem a combination of two active ingredients. The other four were closed by taking the structure from ChEMBL, see <code>docs/06</code> §2.4                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>PERF-02</b> | Code coverage of the verification benches                        | <b>partially measured</b> : 49.89 % of statements over the widest path ( <code>audit_copertura</code> ). Whole-battery coverage and branch coverage remain unmeasured; see §8                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>UI-07</b>   | Behaviour on Firefox and WebKit                                  | <b>not verified</b> — the battery runs on Chromium only; see §8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

## 7. Overall coverage

| Category                   | Requirements | Verified by a bench | Coverage     |
|----------------------------|--------------|---------------------|--------------|
| Scientific (SCI-01...34)   | 33           | 33                  | 100 %        |
| AI agent (AI-01...10)      | 10           | 10                  | 100 %        |
| Stability (STA-01...08)    | 8            | 8                   | 100 %        |
| Interface (UI-01...11)     | 9            | 9                   | 100 %        |
| Security (SEC-01...07)     | 5            | 5                   | 100 %        |
| Architecture (ING-01...01) | 1            | 1                   | 100 %        |
| <b>Automated total</b>     | <b>66</b>    | <b>66</b>           | <b>100 %</b> |
| Not automated (§6)         | 5            | 0                   | 0 %          |
| <b>Declared total</b>      | <b>71</b>    | <b>66</b>           | <b>93 %</b>  |

Coverage is computed over the requirements **declared in this document** and must not be confused with code coverage: it measures how many requirements have a bench verifying them, not how many lines are executed during the tests.

**Why two totals.** Reporting only the 100 % of automated requirements would be true and misleading at once: it is 100 % of what was *chosen* to be automated. The number that matters to an assessor is the second one — **54 requirements verified out of 59 declared** — and the five that are missing are listed by name in §6, not summarised into a percentage.

This table is checked by `tools/verifica-documenti.js`, which counts the identifiers actually present in the document and fails if the totals written here do not match. It is a check born of a real error: the totals said 36 over 9 scientific categories while the identifiers defined were 10, and `SCI-10` appeared twice — once as a verified requirement and once as an unverified one.

## 8. Declared coverage gaps

These are not defects: they are things that have **not been measured**, and that therefore cannot be asserted.

| Gap                                  | What it entails                                                                                                                                            | Why it is so                                                                                                                                                                                                                                                         |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Partial code coverage</b>         | Measured: <b>49.89 %</b> of statements over the widest path a bench walks. It is not the coverage of the whole battery, and it is statements, not branches | Chromium's profiler collects it inside the engine, with no build and without rewriting the source: the obstacle belonged to the tool ( <code>c8</code> , <code>istanbul</code> ), not to the problem. Value recorded; <code>audit_copertura</code> fails if it falls |
| <b>Chromium only</b>                 | Behaviour on Firefox and WebKit is verified by hand, not by a bench                                                                                        | The battery uses <code>playwright-core</code> , which downloads a single engine. In the verification environment the CDN for the other engines answers <b>403</b> to the network policy: they cannot be installed there                                              |
| <b>No visual regression</b>          | An unintended graphical change would not be caught                                                                                                         | Spectra are deterministic and comparable byte for byte ( <code>SCI-05</code> ): the comparison exists on the traces, not on the whole page                                                                                                                           |
| <b>2 entries without a structure</b> | Two entries out of 178 have no structure to show: Ivermectin is a mixture of homologues, Coartem a combination of two active ingredients. It was six       | See <code>06-Scientific-Accuracy-Data-Provenance.md</code> §2.4                                                                                                                                                                                                      |
| <b>Performance on slow devices</b>   | First-paint time is not measured on low-end hardware                                                                                                       | It requires physical devices; the test is manual                                                                                                                                                                                                                     |

\_Document updated to version `bsi-v192`.\_